Zhijun He

Email: hezhj56@mail2.sysu.edu.cn | Tel: (+86)18851059206

EDUCATION

School of Biosafety, Sun Yat-sen University, Shenzhen, China **Sep 2024 - July 2027**

Master of Social Medicine and Health Management

- Average Score: 92.5 / 100 Ranking: 1 / 75 (**Top 1%**)
- Core Courses: Mathematical Statistics (99), Medical Informatics Programming (95.2), AI in Public Health (97.8)

School of Economics and Management, Nanjing Agricultural University, Nanjing, China **Sep 2020 - July 2024**

Bachelor of Agricultural Economy and Management

- Average Score: 91.2 / 100 Ranking: 3 / 93 (**Top 3%**)
- Core Courses: Advanced Mathematics (96), Python Programming (95), Multivariate Statistics (97), Big Data Applications (92)

RESEARCH INTERESTS

Artificial Intelligence in Healthcare, Big Data Medical Research, Digital Health, Healthy Aging

PUBLICATIONS

- **He, Z.**, Han, J., Qiao, X., Zhan, Y., & Xu, M.. (2026). Phenotype identification and precise intervention for multimorbidity of rheumatoid arthritis and diabetes mellitus using interpretable machine learning. *Digital Health*, 12. (**JCR Q1**)
- **He, Z.**, Pei, X., Li, X., Zhao, Y., & Xu, M.. (2025). Uncovering multimorbidity patterns linked to disability in aging populations: A machine learning analysis. *Geriatrics & Gerontology International,* 25, 1200–1208. (**JCR Q2**)
- **He, Z.**, Han, J., Huang, X., & Xu, M.. (2026). The impact of dynamic income disparities on disability progression in older adults: A joint group-based trajectory modeling study. *Journal of Applied Gerontology*, 45(5), 1045–1054. (**JCR Q2**)
- Wu, J., Zheng, Y., **He, Z.**, Zuo, Q., Zhang, W., Li, P., & Zhang, L.. (2026). Leveraging time-series electronic health records with large language models for chronic kidney disease diagnosis in primary care. *BMJ Health Care Inform*. (**Accepted, JCR Q1**)

ACADEMIC PAPERS

- **He, Z.**, & Xu, M.. (2026). Habit-constrained dynamic optimization of plant-based dietary strategies for disability prevention in older adults: A deep reinforcement learning study. *Communications Health*. (**Under Review, JCR Q1**)
- **He, Z.**, Pei, X., Liu, S., Yang, W., & Xu, M.. (2026). Mapping how population-based studies measure disability rates: An AI-assisted scoping review. *The Lancet Healthy Longevity*. (**Submitted, JCR Q1**)

RESEARCH EXPERIENCE

Dynamic Disability Risk Trajectory Prediction Based on Multi-Source Heterogeneous Data **Oct 2025 - Present**

Leader, Sun Yat-sen University & National Health Science Research Center

- **Feature Encoding:** Cleaned and extracted data from over 15 million EHR records, and developed specific encoders for deep representation learning of multi-source heterogeneous features (static, temporal, textual, and discrete).
- **Embedding Fusion:** Developed innovative strategies for fusing representation vectors, including gated networks for adaptive weighting and multi-head attention for deep interaction, ultimately generating personalized representation sequences.
- **Dynamic Disability Prediction:** Using Transformer models with multi-layer self-attention to capture long-term dependencies in representation sequences, outputing multidimensional disability trajectories (ADL, cognition, motor, emotion, and sensation).

Leveraging Large Language Models on Time-series EHR Data for Precise CKD Diagnosis **July 2025 - Sep 2025**

Summer Research Internship, Advanced Institute of Information Technology, Peking University

- **LLM-based CKD Diagnosis:** Developed a DeepSeek LLM-based CKD diagnosis framework using time-series EHR data, where prompt engineering was used to guide binary classification and probability scoring, enabling early CKD identification in primary care settings.

- **Model Evaluation and Comparison:** Comparative experiments using different EHR time series showed that one-month data achieved the best diagnostic accuracy (AUC=0.921), while one-year data yielded higher sensitivity (0.922). The LLM-based model outperformed traditional ML models (GBDT, LightGBM, RF, SVM and LR) across all metrics.
- **Research Outcomes:** Contributed to LLM prompt design, model tuning, diagnostic output and evaluation, and co-authored the manuscript, which has been accepted by *BMJ Health & Care Informatics*.

Dynamic Optimization of Dietary Strategies for Disability Prevention in Older Adults **July 2025 - Apr 2026**

Leader, Sun Yat-sen University

- **Personalized dietary strategies:** For the limitation of traditional static association analysis, we employed deep reinforcement learning to construct personalized plant-based dietary strategies, achieving dietary adjustments delaying disability in older adults.
- **Model Performance:** Using the CLHLS longitudinal data (2008-2018), we found that model-recommended dietary strategies were associated with lower predicted disability burden and a higher likelihood of disability-free trajectories among older adults.
- **Research Outcomes:** Contributed to the entire process of data collection, algorithm development and model validation, and prepared the article as first author and submitted to *Communications Health*.

Analysis of Multimorbidity Patterns in Older Adults and Specific Multimorbidity Phenotypes **Oct 2024 - Dec 2025**

Leader, Sun Yat-sen University

- **Multimorbidity Patterns:** Uncovered 10 multimorbidity patterns among older adults in Shenzhen using self-organizing maps and K-means. Logistic regression was applied to analyze associations between multimorbidity patterns and disability.
- **Multimorbidity Phenotypes:** Developed the machine learning models to classify the multimorbidity status of rheumatoid arthritis and diabetes mellitus. The interpretable SHAP values were clustered to identify four distinct multimorbidity phenotypes.
- **Research Outcomes:** As the primary contributor to data analysis and article writing, and the research findings were published in *Digital Health* and *Geriatrics & Gerontology International* respectively.

Association Between Dynamic Income Disparity and Disability Progression in Older Adults **July 2023 - July 2024**

Leader, Sun Yat-sen University

- **Trajectory Fitting:** Using CLHLS longitudinal data (2002-2018), we applied group-based trajectory modeling to construct joint trajectories of income and disability in older adults, and calculated conditional probabilities between trajectories.
- **Association Analysis:** Multinomial logistic regression was used to explore the association between income trajectories and disability trajectories, and the association differences across age groups were further compared.
- **Research Outcomes:** Independent modeling and analysis, with wrote article published in *Journal of Applied Gerontology*.

CONFERENCE EXPERIENCE

The 2nd International Forum on Proactive Health and Aging: AI-Empowered Disease Management (Poster) **August 2026**

The 5th Huaxia Public Health Forum: Public Health Advancement Drived by Digital Intelligence (Oral Report) **July 2025**

AWARDS & HONORS

- National Scholarship (Top 0.2%, 2025)
- National Encouragement Scholarship (Top 3%, 2023)
- Outstanding Undergraduate Thesis, Nanjing Agricultural University (Top 5%, 2024)
- Excellent Graduate, Nanjing Agricultural University (Top 10%, 2024)
- First Prize Graduate Scholarship, Sun Yat-sen University (2024, 2025)
- The Rongchuang Foundation Scholarship, Sun Yat-sen University (2025)
- Second Prize in the National Mathematics Competition and Elite Challenge (2025)

SKILLS

- Professional Tools: Python, R Studio, Stata, SPSS, Origin, MS Office
- Algorithm skills: Machine learning, Deep learning, Reinforcement learning, LLM-based workflow
- Language: English (proficient), Chinese (native)